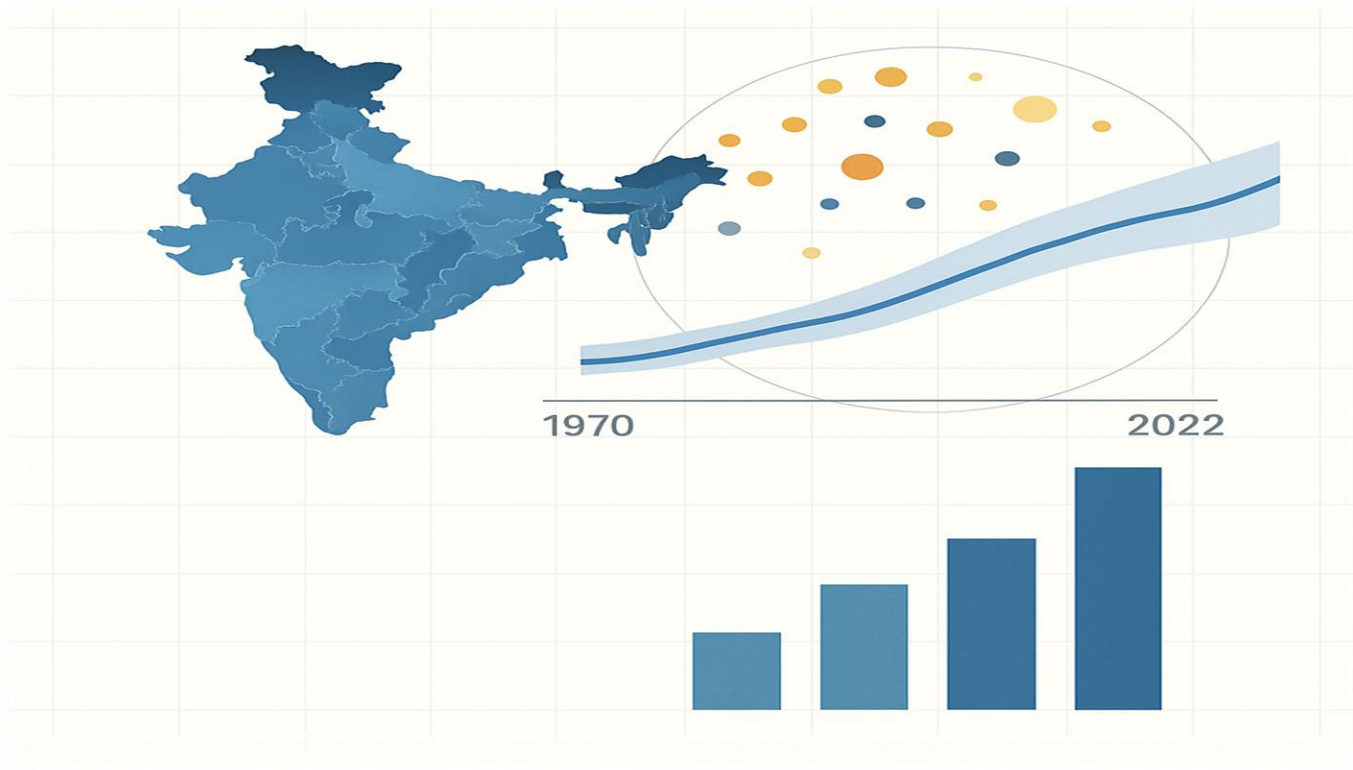
PROJECT ON RELATIONSHIP BETWEEN POPULATION DENSITY AND LIFE EXPECTANCY OF INDIA (1970-2022)



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DEPARTMENT OF STATISTICS

CONTENTS: -



- DATA SOURCE
- DATASETS TO BE ANALYZE
- INITIAL DATA VISUALIZATION AND INTERPRETATION
  - Plot population density (1970-2022)
  - Plot life expectancy (1970-2022)
- DATA PREPARATION AND TRANSFORMATION
  - Stationary test by ADF
  - Log Transformation and Differencing
  - Plot of Transformed Series
- AUTOCORRELATION ANALYSIS AND MODEL BUILDING
  - ACF and PACF Plots for Transformed Series
  - Model identification
- FORECASTING POPULATION DENSITY
  - ARIMA Model Forecast
  - Holt's Exponential Smoothing Forecast
  - Comparison and Interpretation
- FORECASTING LIFE EXPECTANCY
  - ARIMA Model Forecast
  - Holt's Exponential Smoothing Forecast
  - Comparison and Interpretation
- MODEL COMPARISION AND FORECAST ACCURACY
  - Metrices used
  - Interpretation
- CONCLUSION
- REFERENCE
- APPENDIX

DATA SOURCE

*The dataset used in this project is sourced from publicly available demographic statistics pertaining to India. It contains annual data on **Population Density** and **Life Expectancy**, spanning from 1970 to 2022.*

Source: World Bank Open Data , Kaggle

Dataset Name: India Population (1970-2022)

Link:

🔗 https://databank.worldbank.org/reports.aspx?source=2&country=IND

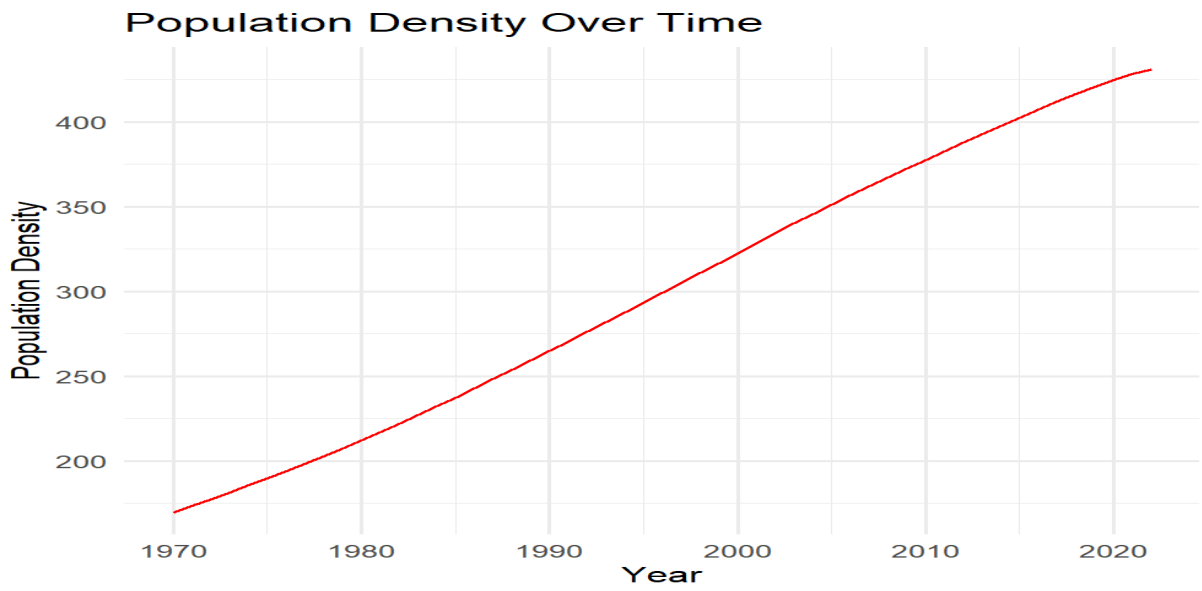
🔗 https://www.kaggle.com/datasets/amritharj/population-of-india-19502022?

DATASETS TO BE ANALYZE

Year	Population Density	Life Expectancy
1970	169.59	47.41
1971	173.4	48.06
1972	177.3	48.72
1973	181.34	49.37
1974	185.48	50
1975	189.68	50.63
1976	193.92	51.25
1977	198.25	51.88
1978	202.68	52.51
1979	207.24	52.99
1980	211.98	53.47
1981	216.86	53.95
1982	221.82	54.43
1983	226.88	54.91
1984	232.08	55.27
1985	237.35	55.62
1986	242.72	55.98
1987	248.14	56.33
1988	253.62	56.69
1989	259.19	57.18
1990	264.8	57.66
1991	270.42	58.15
1992	276.09	58.63
1993	281.8	59.12
1994	287.55	59.59
1995	293.34	60.06
1996	299.12	60.53
1997	304.92	61
1998	310.73	61.47
1999	316.53	61.88
2000	322.35	62.28
2001	328.23	62.69
2002	334.11	63.09
2003	339.92	63.5
2004	345.66	63.91
2005	351.25	64.31
2006	356.64	64.72
2007	361.91	65.12
2008	367.09	65.53
2009	372.24	65.98
2010	377.4	66.43
2011	382.57	66.87
2012	387.71	67.32
2013	392.77	67.77
2014	397.67	68.07
2015	402.42	68.37
2016	407.22	68.67
2017	411.95	68.97
2018	416.46	69.27
2019	420.75	69.5
2020	424.79	69.73
2021	428.19	69.96
2022	431.11	70.19

INITIAL DATA VISUALIZATION AND INTERPRETATION

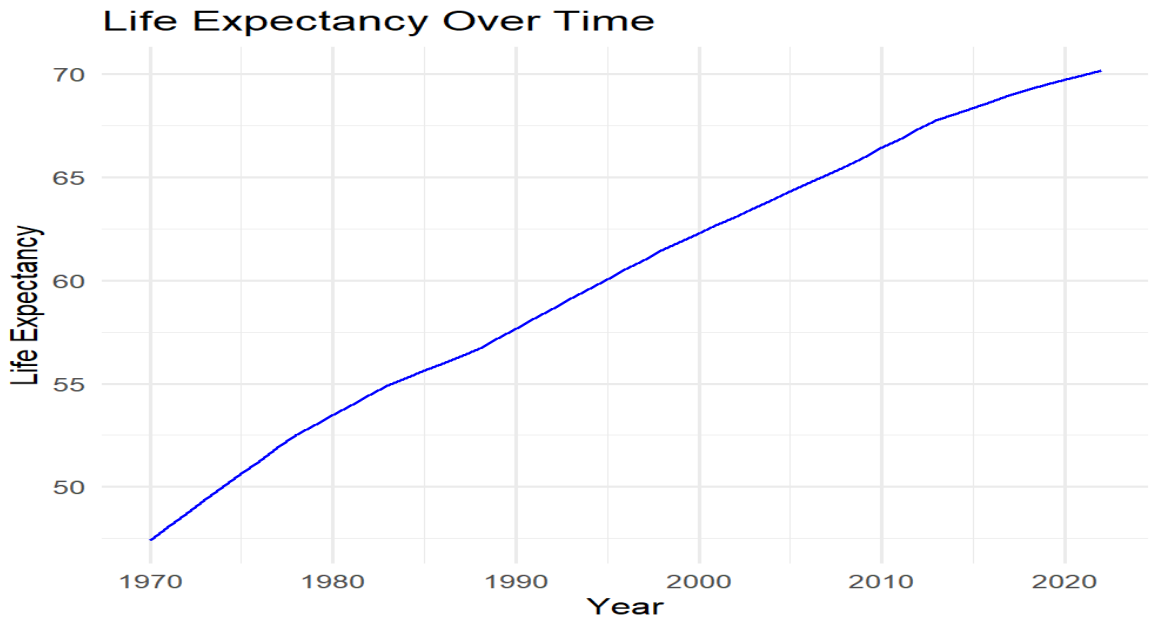
📉 PLOT OF POPULATION DENSITY OVER TIME:



Population Density Over Time:

- **Steady Increase:** Population density has steadily increased over the 50-year period.
- **Possible Acceleration:** The rate of increase seems to **accelerate** in the later years (post-2000). This could indicate rapid urbanization or other demographic shifts.

PLOT OF LIFE EXPECTANCY OVER TIME



Life Expectancy Over Time:

- **Significant Improvement:** Life expectancy has shown a **significant improvement** over the 50-year period.
- **Slower Initial Growth:** The growth in life expectancy appears to be **slower** in the early years (1970s and 1980s).
- **Faster Recent Growth:** The growth rate seems to **increase** in the later years, suggesting advancements in healthcare, living standards, and possibly, better access to resources

.

INTERPRETATION: -

- **It is clear that from the two plots, it can be thought that the random fluctuations as well as cyclic fluctuations are almost NIL in both population density as well as life expectancy.**
- **Moreover, for both of the cases an upward almost linear trend line is visible.**

DATA PREPARATION AND TRANSFORMATION

- ✓ **TO CHECK STATIONARY OF POPULATION DENSITY WITH TIME BY ADF TEST:-**

Augmented Dickey-Fuller Test

data: y_population
Dickey-Fuller = 0.91259, Lag order = 3, p-value = 0.99
alternative hypothesis: stationary
.

- ✓ **TO CHECK STATIONARY OF LIFE EXPECTANCY WITH TIME BY ADF TEST:-**

Augmented Dickey-Fuller Test

data: y_life_expectancy
Dickey-Fuller = -0.35719, Lag order = 3, p-value = 0.9849
alternative hypothesis: stationary

INTERPRETATION: -

- **We know if p value <0.05 then the plot of time series is said to be stationary. HERE FOR BOTH THE CASES, p value >0.05 which means the series are non-stationary.**

- **Now we move forward to check the difference whether it is <0.05.**

✓ **FOR POPULATION DENSITY-**

Augmented Dickey-Fuller Test

data: y_population_diff
Dickey-Fuller = 1.4254, Lag order = 3, p-value = 0.99
alternative hypothesis: stationary

✓ **FOR LIFE EXPECTANCY-**

Augmented Dickey-Fuller Test

data: y_life_expectancy_diff
Dickey-Fuller = -2.3346, Lag order = 3, p-value = 0.4398
alternative hypothesis: stationary

INTERPRETATION: -

- **Subsequently, differencing was applied: the ADF test results showed p-values >0.05 for both series, after differencing at order 1 indicating that they are non-stationary.**

Now we perform log transformation to see p value<0.05.

✓ **APPLYING LOG TRANSFORMATION FOR POPULATION DENSITY-**

Augmented Dickey-Fuller Test

data: log_population
Dickey-Fuller = 1.9422, Lag order = 3, p-value = 0.99
alternative hypothesis: stationary

✓ **APPLYING LOG TRANSFORMATION FOR LIFE EXPECTANCY-**

Augmented Dickey-Fuller Test

data: log_life_expectancy
Dickey-Fuller = -0.46766, Lag order = 3, p-value = 0.98
alternative hypothesis: stationary

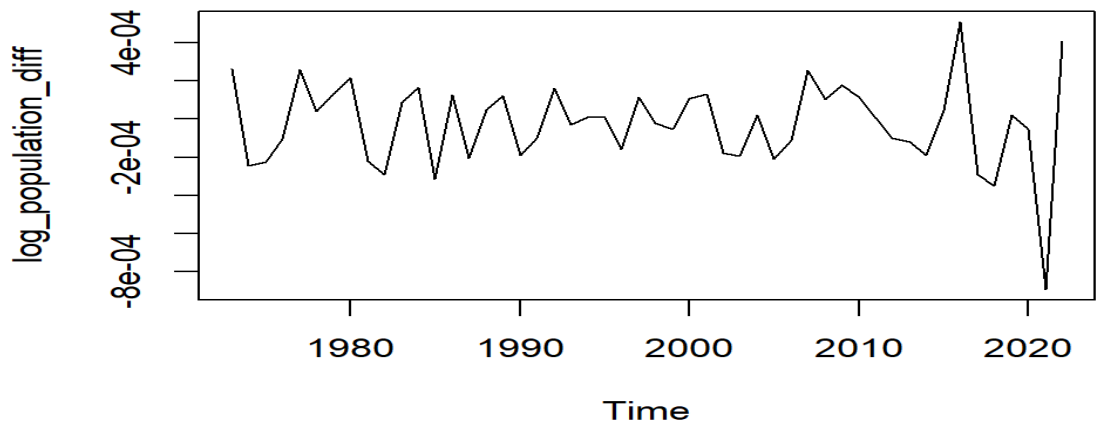
INTERPRETATION: -

- **After performing the Augmented Dickey-Fuller (ADF) test on both the original and log-transformed time series data for *Population Density* and *Life Expectancy*, it was observed that the results remained consistent. Specifically, both the original and log-transformed series were found to be non-stationary, requiring differencing to achieve stationarity.**

Further differencing was still necessary to proceed with ARIMA modelling and forecasting.

✓ **PLOT OF TRANSFORMED SERIES**

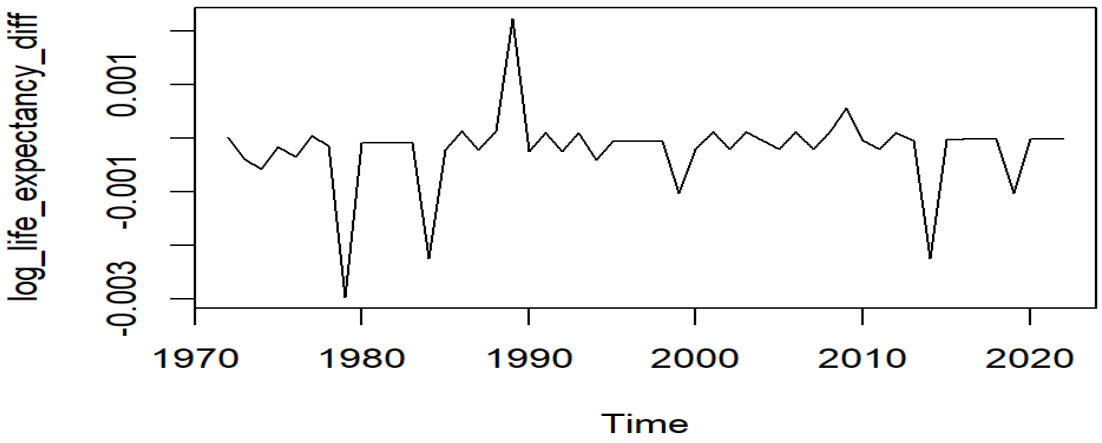
PLOT OF 3rd ORDER DIFFERENCE OF POPULATION DENSITY



Augmented Dickey-Fuller Test

data: log_population_diff
Dickey-Fuller = -3.9187, Lag order = 3, p-value = 0.02021
alternative hypothesis: stationary

PLOT OF 2nd ORDER DIFFERENCE OF LIFE EXPECTANCY-



Augmented Dickey-Fuller Test

data: log_life_expectancy_diff
Dickey-Fuller = -3.564, Lag order = 3, p-value = 0.04467
alternative hypothesis: stationary

INTERPRETATION:

Subsequently, differencing was applied:

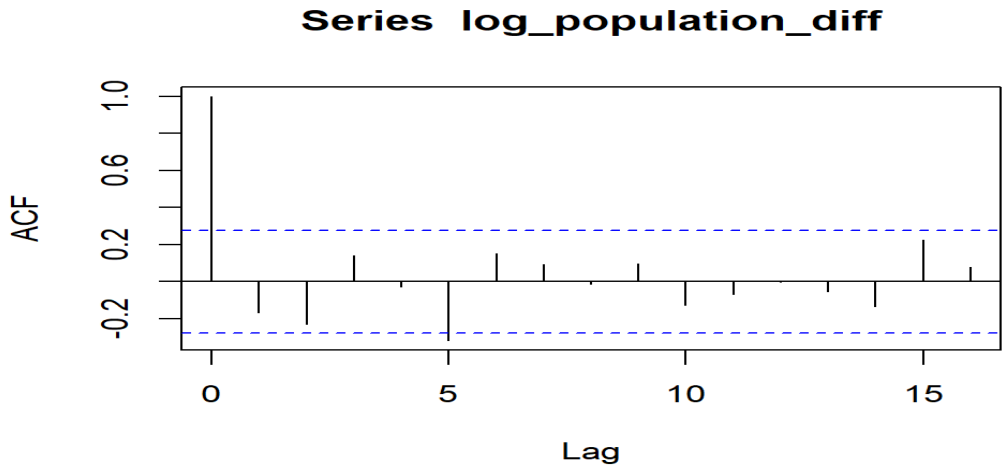
- **A third-order difference was applied to the log-transformed *Population Density* series.**
- **A second-order difference was applied to the log-transformed *Life Expectancy* series.**

After differencing, the ADF test results showed p-values < 0.05 for both series, indicating that they became stationary. This confirms that the appropriate level of differencing successfully removed non-stationarity from the data, making the series suitable for time series modelling using ARIMA and other forecasting techniques.

Now, we analysed the ACF and PACF plots to identify the nature of the autoregressive and moving average components. These plots help in identifying the lag structures and in model selection for ARIMA.

AUTOCORRELATION ANALYSIS

✓ **ACF Plot of population density:-**

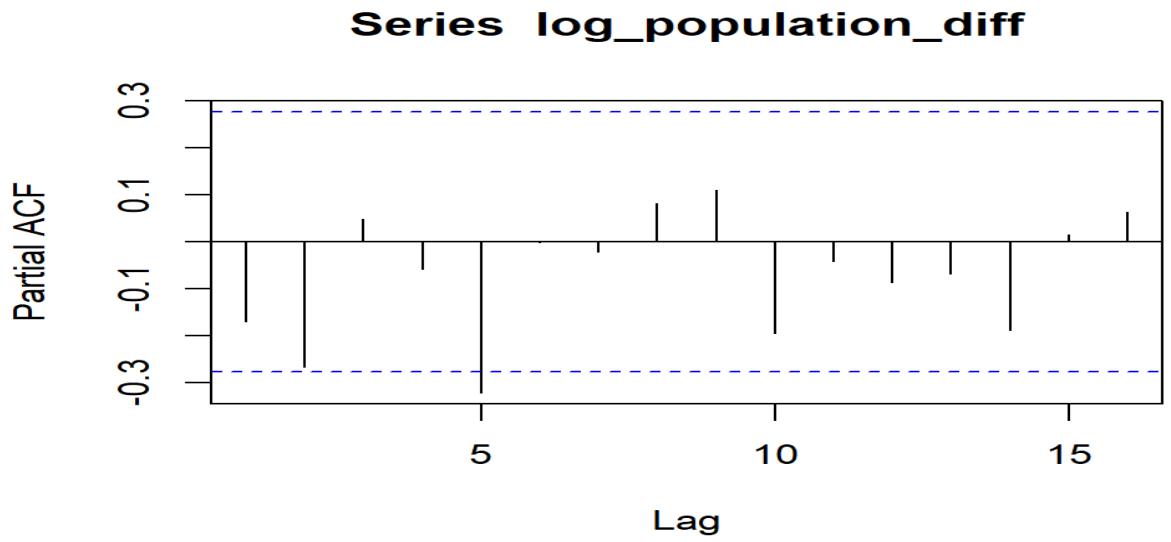


Autocorrelations of series 'log_population_diff', by lag

0	1	2	3	4	5	6	7	8	9	10
1.000	-0.170	-0.230	0.139	-0.029	-0.319	0.152	0.092	-0.015	0.098	-0.127
11	12	13	14	15	16					
-0.069	-0.002	-0.054	-0.137	0.224	0.077					

The rapid decay after lag 0 suggests that a low-order autoregressive model (e.g., AR(1)) may be suitable for modeling the series.

✓ PACF Plot of population density: -



Partial autocorrelations of series 'log_population_diff', by lag

1	2	3	4	5	6	7	8	9	10	11
-0.170	-0.267	0.049	-0.058	-0.321	-0.001	-0.021	0.082	0.111	-0.195	-0.042

12	13	14	15	16
-0.087	-0.068	-0.188	0.016	0.064

No spikes clearly exceed the confidence bounds — some come close (e.g., lag 5), but they remain within the range. This suggests that partial autocorrelations are not significant at most lags.

✓ MODEL BUILDING-

```
Series: y_population
ARIMA(1,2,0)

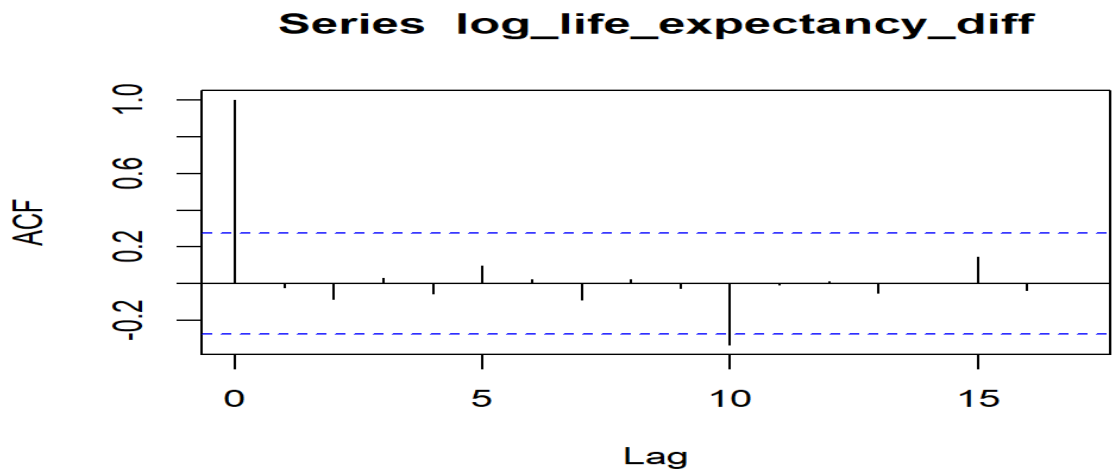
Coefficients:
         ar1
      0.9100
s.e.  0.0703

sigma^2 = 0.007773:  log likelihood = 54.91
AIC=-105.81   AICc=-105.56   BIC=-101.95
```

The mathematical model of this series is

$$\Delta^2 Y_t = O.9100\ \Delta^2 Y_{t-1} + Z_t$$

✓ ACF Plot of life expectancy: -



Autocorrelations of series 'log_life_expectancy_diff', by lag

0	1	2	3	4	5	6	7	8	9	10
1.000	-0.020	-0.085	0.030	-0.054	0.097	0.022	-0.090	0.024	-0.027	-0.333
11	12	13	14	15	16	17				
-0.007	0.013	-0.050	0.002	0.147	-0.037	0.002				

ACF: Spike at lag 0 and lag 10 is significant (outside blue line) → Possible MA(1)

✓ *PACF Plot of life expectancy: -*



Partial autocorrelations of series 'log_life_expectancy_diff', by lag

1	2	3	4	5	6	7	8	9	10	11
-0.020	-0.085	0.027	-0.061	0.101	0.014	-0.070	0.016	-0.031	-0.343	-0.041

12	13	14	15	16	17
-0.030	-0.054	-0.040	0.247	-0.026	-0.028

PACF: All within the lines (only lag 10 is significant) → No strong AR component.

✓ **MODEL BUILDING:**

```
Series: y_life_expectancy
ARIMA(0,2,0)

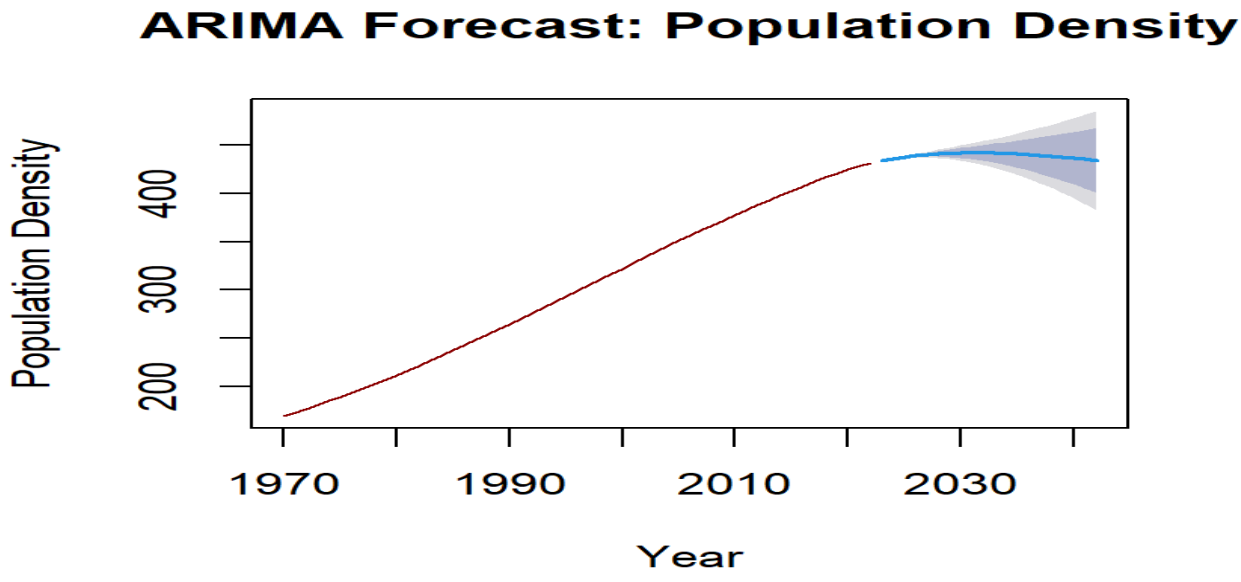
sigma^2 = 0.001838:  log likelihood = 89.47
AIC=-176.94   AICc=-176.86   BIC=-175.01
```

The mathematical model of this series is

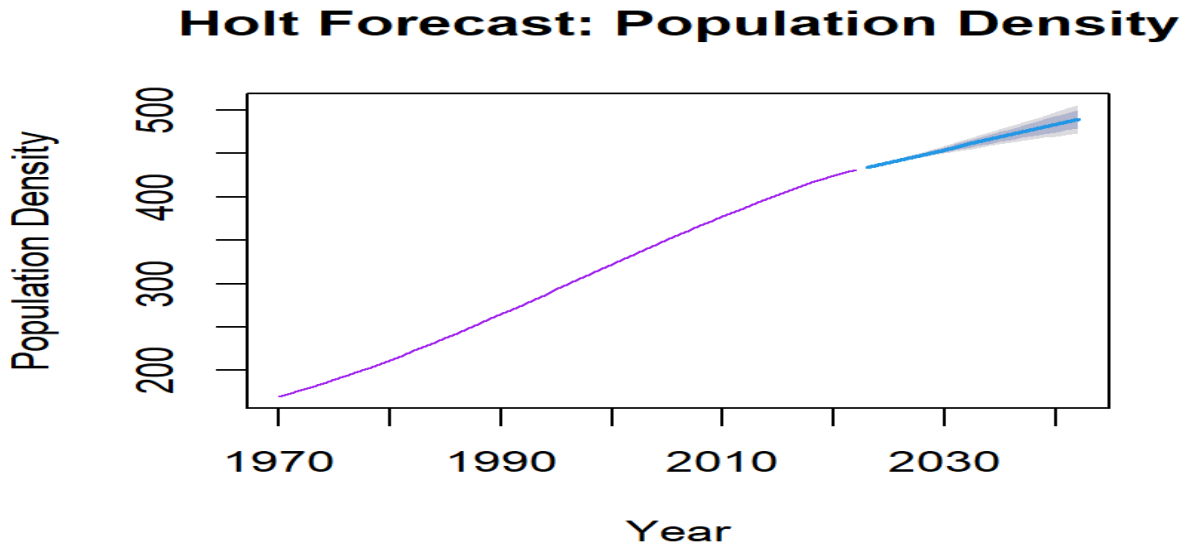
$$\Delta^2 Y_t = +Z_t$$

FORECASTING POPULATION DENSITY

Forecast of Population Density Using ARIMA



Holt's Forecast for Population Density

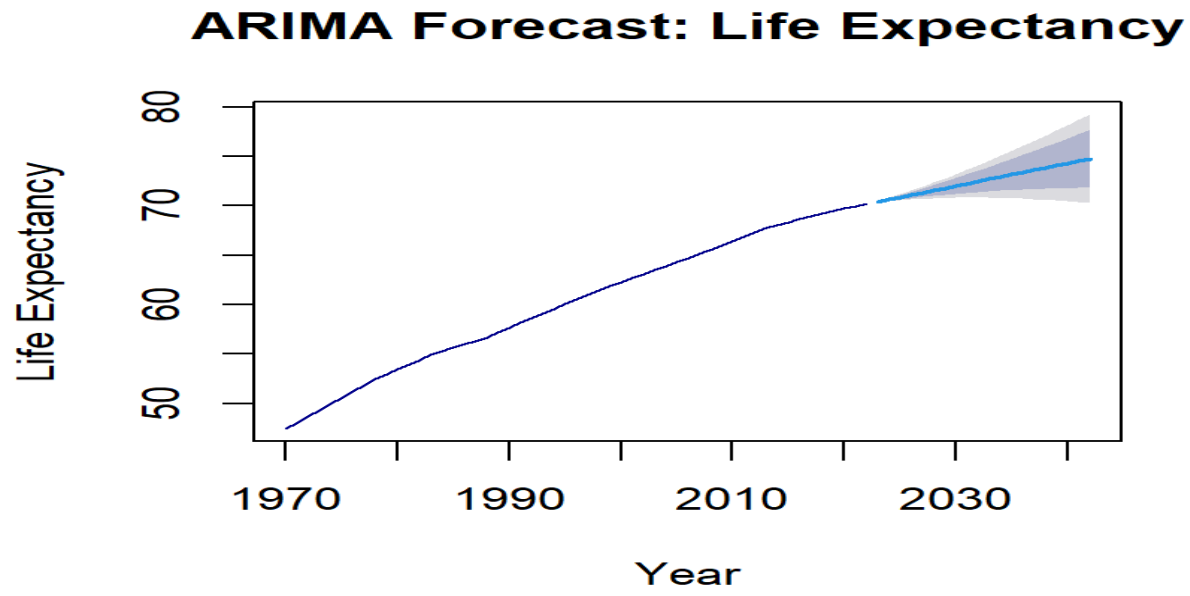


✓ **Now we compare forecast of population density over time for next 20 years by auto. arima and Holt's method: -**

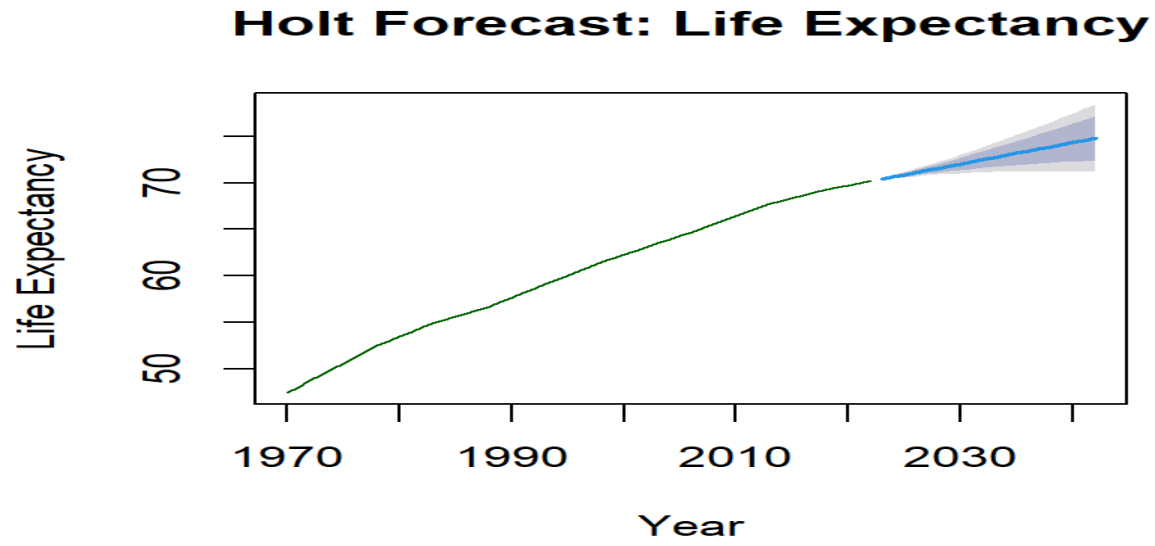
YEAR	Forecast by auto.arima	Forecast by Holt's method
2023	433.5932	434.0300
2024	435.6789	436.9500
2025	437.4029	439.8700
2026	438.7978	442.7900
2027	439.8931	445.7100
2028	440.7158	448.6300
2029	441.2905	451.5499
2030	441.6395	454.4699
2031	441.7831	457.3899
2032	441.7397	460.3099
2033	441.5263	463.2299
2034	441.1581	466.1499
2035	440.6490	469.0698
2036	440.0118	471.9898
2037	439.2579	474.9098
2038	438.3978	477.8298
2039	437.4412	480.7498
2040	436.3967	483.6698
2041	435.2722	486.5898
2042	434.0749	489.5097

FORECASTING LIFE EXPECTANCY

Forecast of Life Expectancy Using ARIMA



Holt's Forecast for Life Expectancy



✓ **Now we compare forecast of life expectancy over time for next 20 years by auto.arima and Holt's method:-**

YEAR	Forecast by auto.arima	Forecast by Holt's method
2023	70.42	70.42019
2024	70.65	70.65039
2025	70.88	70.88058
2026	71.11	71.11077
2027	71.34	71.34097
2028	71.57	71.57116
2029	71.80	71.80135
2030	72.03	72.03155
2031	72.26	72.26174
2032	72.49	72.49193
2033	72..72	72.72213
2034	72.95	72.95232
2035	73.18	73.18251
2036	73.41	73.41271
2037	73.64	73.64290
2038	73.87	73.87309
2039	74.10	74.10329
2040	74.33	74.33348
2041	74.56	74.56367
2042	74.79	74.79387

MODEL COMPARISION AND FORECAST ACCURACY

To evaluate the forecasting performance of the models, we calculated the **Mean Squared Error (MSE)** for both ARIMA and Holt's method on each time series.

Mean Squared Error (MSE) Comparison

Model	Population density	Life Expectancy
ARIMA	0.007333129	0.001768162
Holt's method	0.02221099	0.001790262

INTERPRETATION:

- For **Population Density**, the **ARIMA model** yielded a smaller error value, indicating better forecast accuracy.
- For **Life Expectancy**, the **ARIMA Model** performed better based on MSE comparison.

CONCLUSION

This project undertook a time series analysis of two key demographic indicators of India — Population Density and Life Expectancy — over a defined historical period. Preliminary visualization and statistical testing indicated both series were non-stationary in their original forms. The Augmented Dickey-Fuller (ADF) tests confirmed appropriate transformations.

- *To achieve stationarity, logarithmic transformation followed by differencing was applied. Autocorrelation and partial autocorrelation functions (ACF and PACF) were analysed to identify the appropriate lag structures for ARIMA modelling.*
- *Using the auto.arima() function to generate 20-period ahead forecasts, which indicated a continuing upward trend in both population density and life expectancy — consistent with historical growth patterns in India.*

- *Forecast accuracy was assessed using Mean Squared Error (MSE), and Holt's exponential smoothing method was implemented as a comparative forecasting approach. Both methods yielded coherent and reliable forecasts, although ARIMA models demonstrated slightly better performance in terms of model fit and accuracy.*

At the end, this study demonstrates the applicability and effectiveness of univariate time series modelling — particularly ARIMA — in forecasting demographic trends. For future research, the analysis can be extended by incorporating multivariate models or additional explanatory variables to further enhance predictive performance.

REFERENCE

❖ **FUNDAMENTAL OF STATISTICS VOLUME 2**

BY-

- A.M. GUN
- M.K.GUPTA
- B. DASGUPTA

R PACKAGES TO BE USED

NAME	DESCRIPTION
ggPlot2	create elegant data visualisations using the grammar of graphics.
T Series	Time series analysis and computational finance
Zoo	S3 Infrastructure for regular and irregular time series(Z's ordered observations)
Forecast	Forecasting functions for time series and linear models

APPENDIX

```
z <- read.csv("C:/Users/srinjan sanyal/Documents/population india.csv", header=TRUE)
z
y_population <- ts(z$Population.Density, start = min(z$Year), frequency = 1)
y_life_expectancy <- ts(z$Life.Expectancy, start = min(z$Year), frequency = 1)

# Population Density Plot using ggplot2
ggplot(z, aes(x = Year, y = Population.Density)) +
  geom_line(color = "red") +
  labs(title = "Population Density Over Time", x = "Year", y = "Population Density") +
  theme_minimal()

# Life Expectancy Plot using ggplot2
ggplot(z, aes(x = Year, y = Life.Expectancy)) +
  geom_line(color = "blue") +
  labs(title = "Life Expectancy Over Time", x = "Year", y = "Life Expectancy") +
  theme_minimal()


# ADF Test for Population Density
adf_test1<-adf.test(y_population)
print(adf_test1)
# ADF Test for Life Expectancy
adf_test2<-adf.test(y_life_expectancy)
print(adf_test2)

y_population_diff <- diff(y_population)
y_life_expectancy_diff <- diff(y_life_expectancy)

adf_result_pop_diff <- adf.test(y_population_diff)
adf_result_life_diff <- adf.test(y_life_expectancy_diff)


print(adf_result_pop_diff)
print(adf_result_life_diff)
```

```
# Apply log transformation
log_population <- log(y_population)
log_life_expectancy <- log(y_life_expectancy)
adf.test(log_population)
adf.test(log_life_expectancy)
log_population_diff <- diff(log_population,differences=3)
log_life_expectancy_diff <- diff(log_life_expectancy,differences=2)

plot.ts(log_population_diff)
plot.ts(log_life_expectancy_diff)

adf.test(log_population_diff)
adf.test(log_life_expectancy_diff)

acf(log_population_diff,plot=FALSE)
pacf(log_population_diff,plot=FALSE)
acf(log_population_diff)
pacf(log_population_diff)
acf(log_life_expectancy_diff,plot=FALSE)
pacf(log_life_expectancy_diff,plot=FALSE)
acf(log_life_expectancy_diff)
pacf(log_life_expectancy_diff)
#auto.arima
model_popdensity <- auto.arima(y_population,seasonal = TRUE)
model_popdensity

model_lifeexpectancy <- auto.arima(y_life_expectancy ,seasonal=TRUE)
model_lifeexpectancy

forecast_pop <- forecast(model_popdensity, h=20)
forecast_life <- forecast(model_lifeexpectancy, h=20)

# Plot forecasts
plot(forecast_pop, main="ARIMA Forecast: Population Density", xlab="Year",
     ylab="Population Density", col="darkred")
plot(forecast_life, main="ARIMA Forecast: Life Expectancy", xlab="Year",
     ylab="Life Expectancy", col="darkblue")
```

```r
MSE<-mean((y_population-fitted(forecast_pop))^2)
MSE
MSE<-mean((y_life_expectancy-fitted(forecast_life))^2)
MSE
# Holt's Method for Population Density
holt_pop <- holt(y_population, h=20)
summary(holt_pop)

# Plot the Holt forecast
plot(holt_pop, main="Holt Forecast: Population Density", xlab="Year", ylab="Population Density", col="purple")

# Holt's Method for Life Expectancy
holt_life <- holt(y_life_expectancy, h=20)
summary(holt_life)

# Plot the Holt forecast
plot(holt_life, main="Holt Forecast: Life Expectancy", xlab="Year", ylab="Life Expectancy", col="darkgreen")

MSE <- mean((y_population - fitted(holt_pop))^2)
MSE
MSE<-mean((y_life_expectancy - fitted(holt_life))^2)
MSE
```